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(54) **BALLISTIC SHIELD HARNESS**

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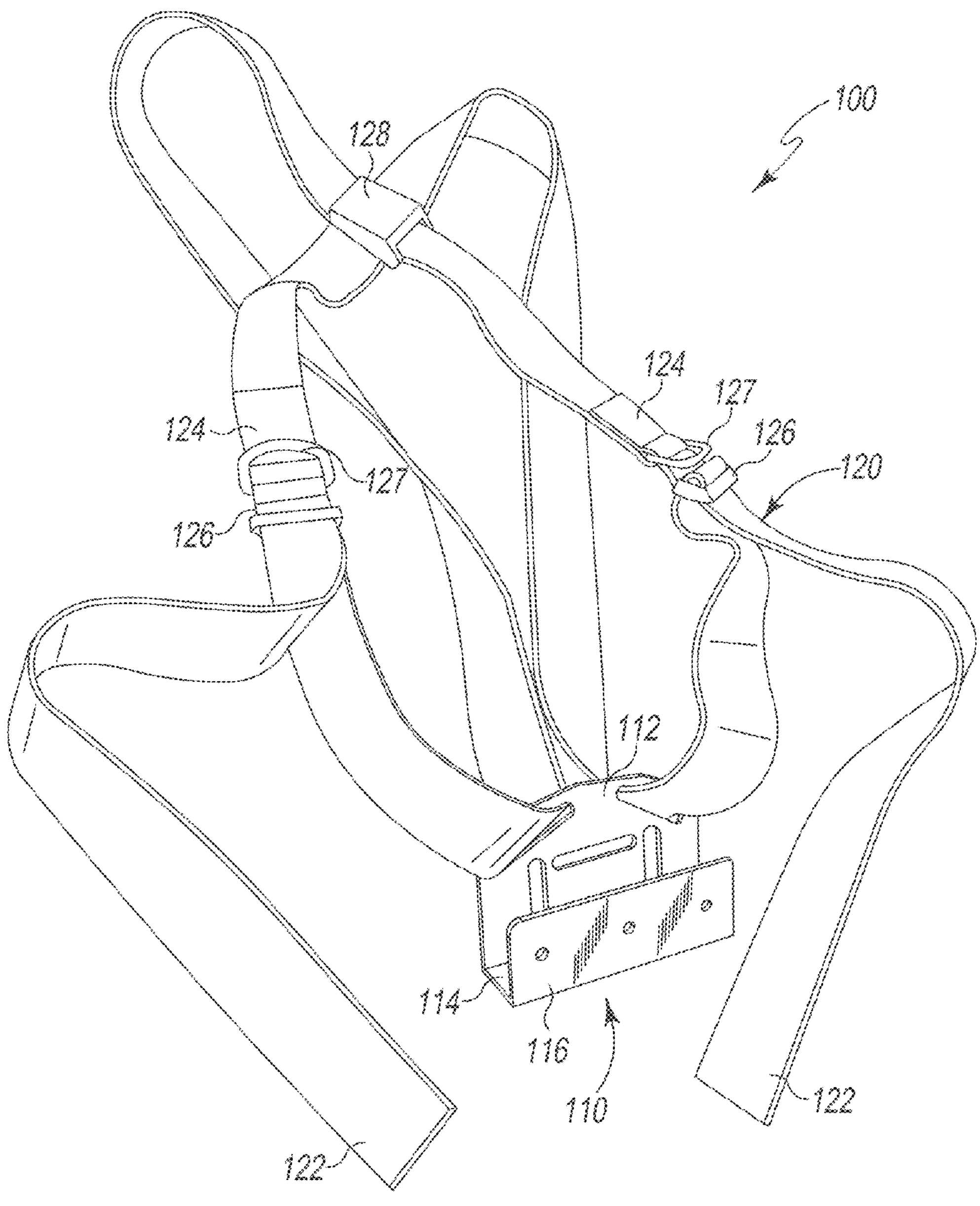
(57) **ABSTRACT**

The shield harness supports the weight of a ballistic shield and distributes the weight of the shield across the user's shoulders and torso. The shield harness includes a U-shaped shield saddle and a pair of adjustable shoulder straps. The shoulder straps wrap over the shoulders of the user to position and support the shield saddle and the weight of a ballistic shield in front of the user.



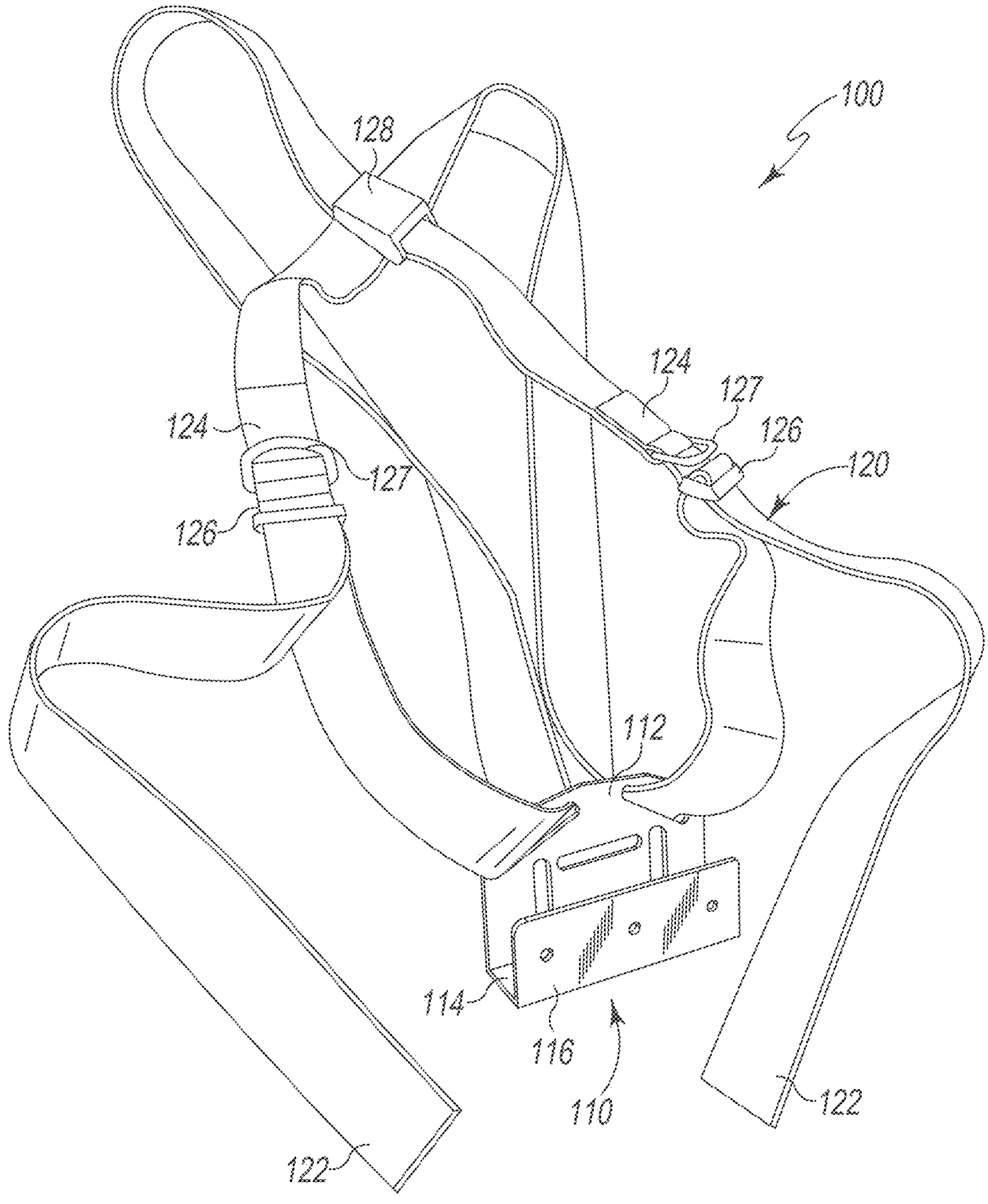


Fig. 1

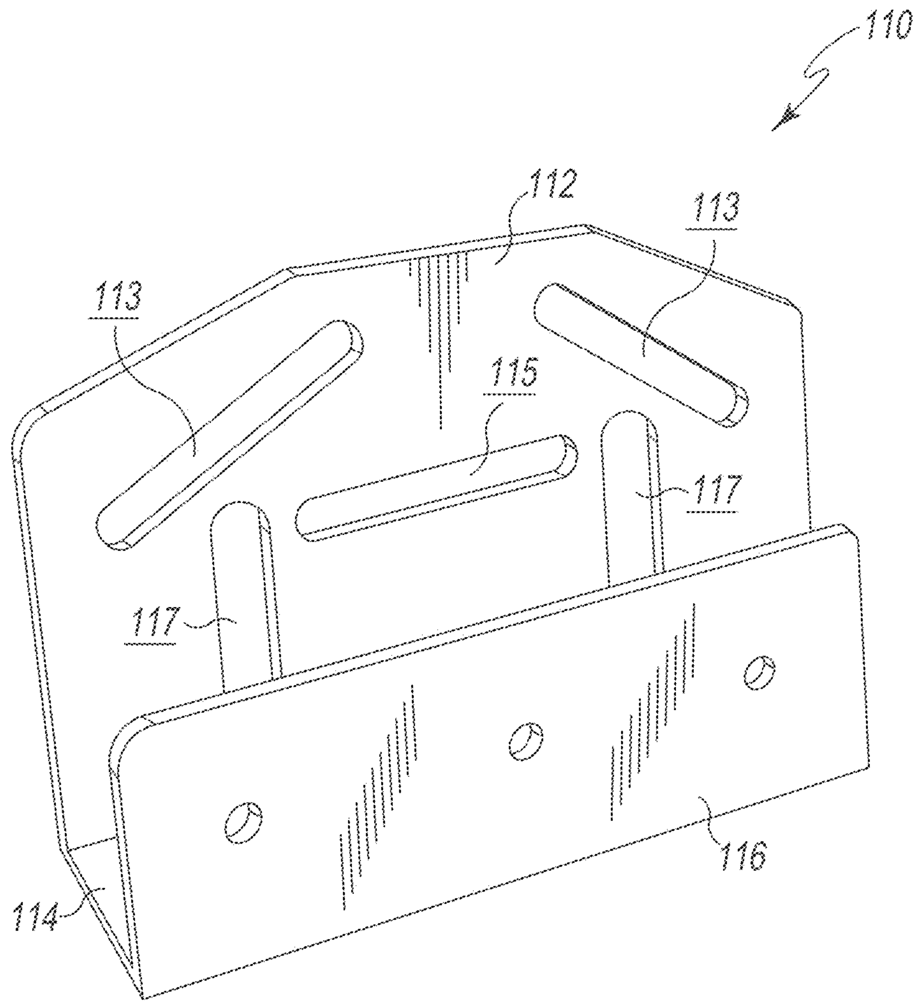


Fig. 2

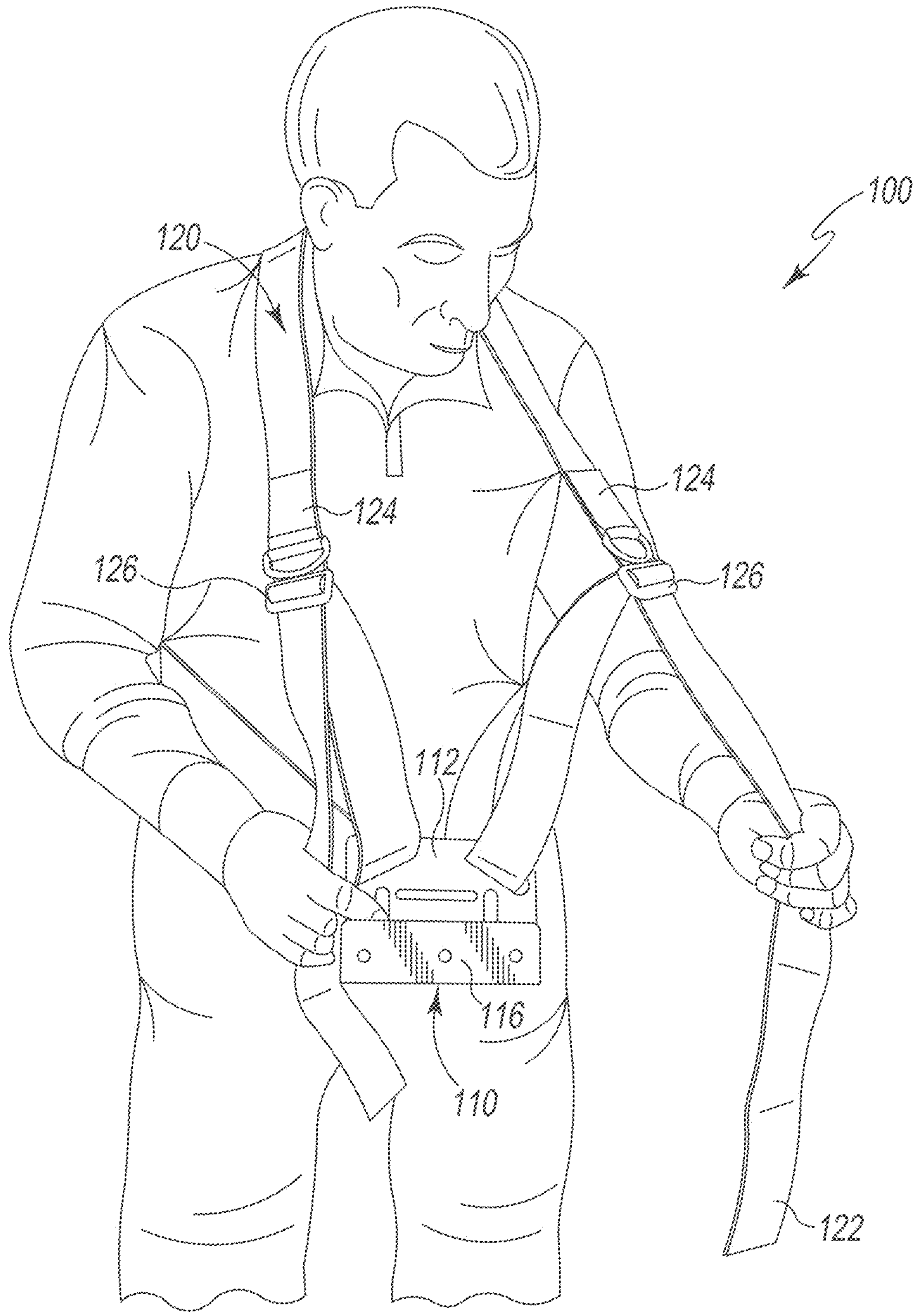


Fig. 3

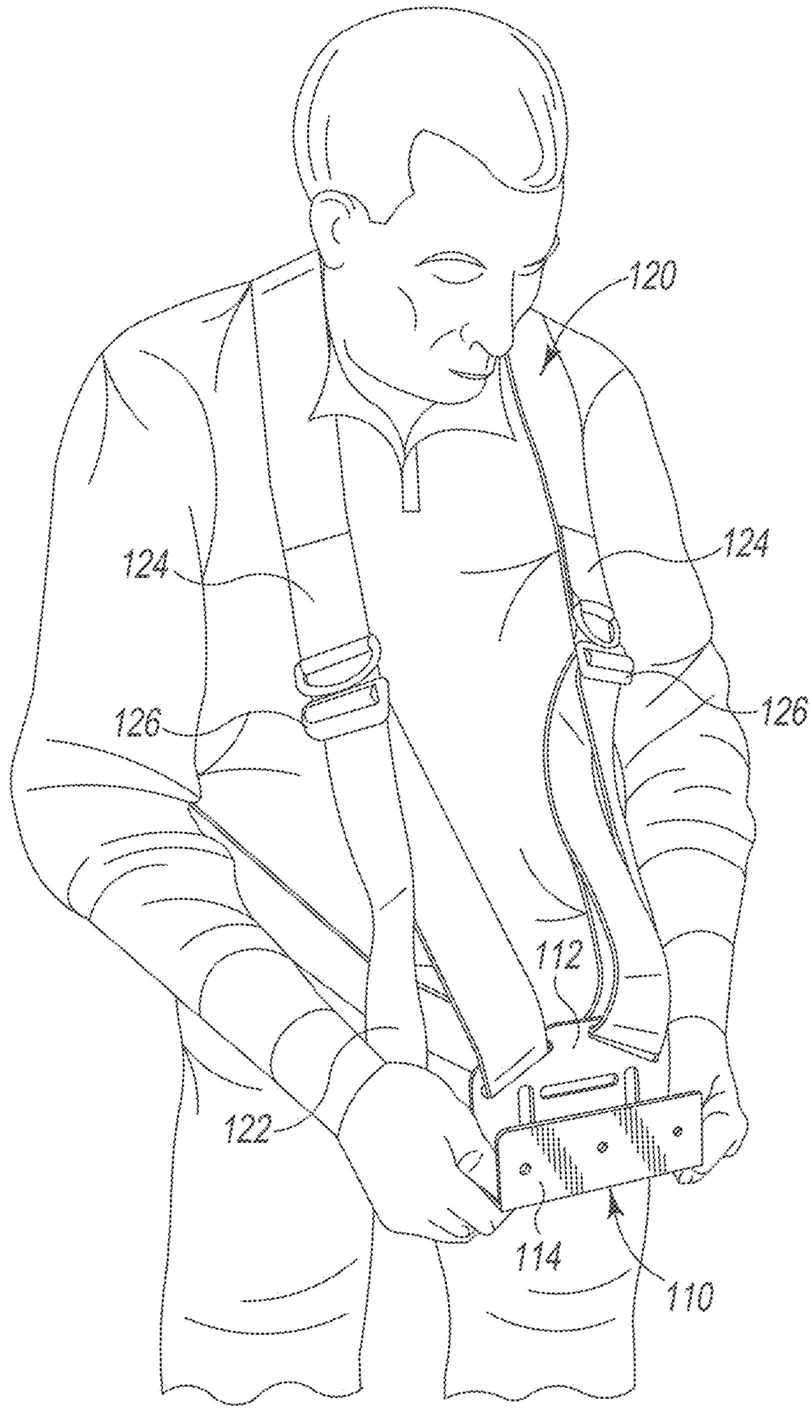


Fig. 4

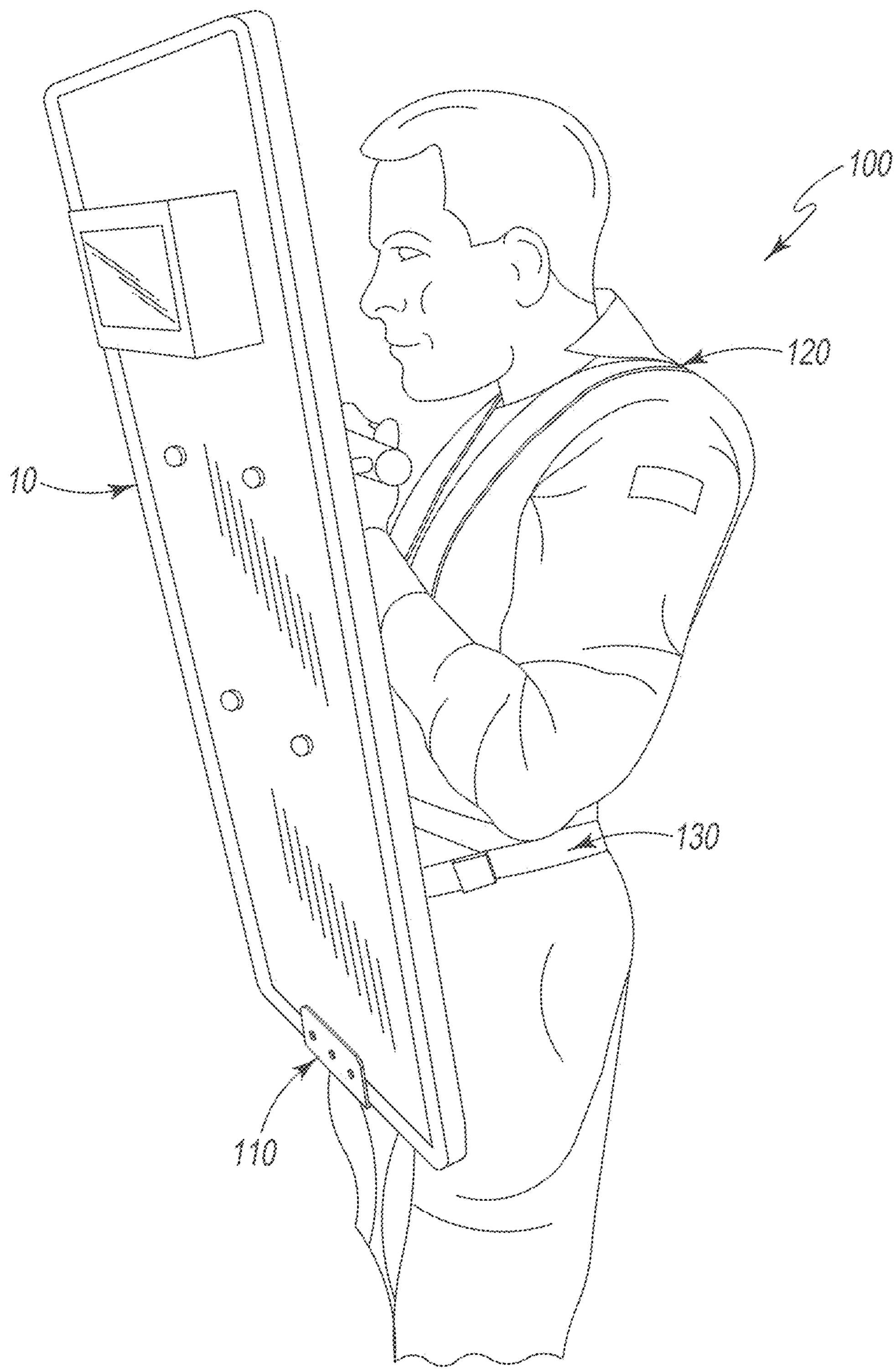


Fig. 5

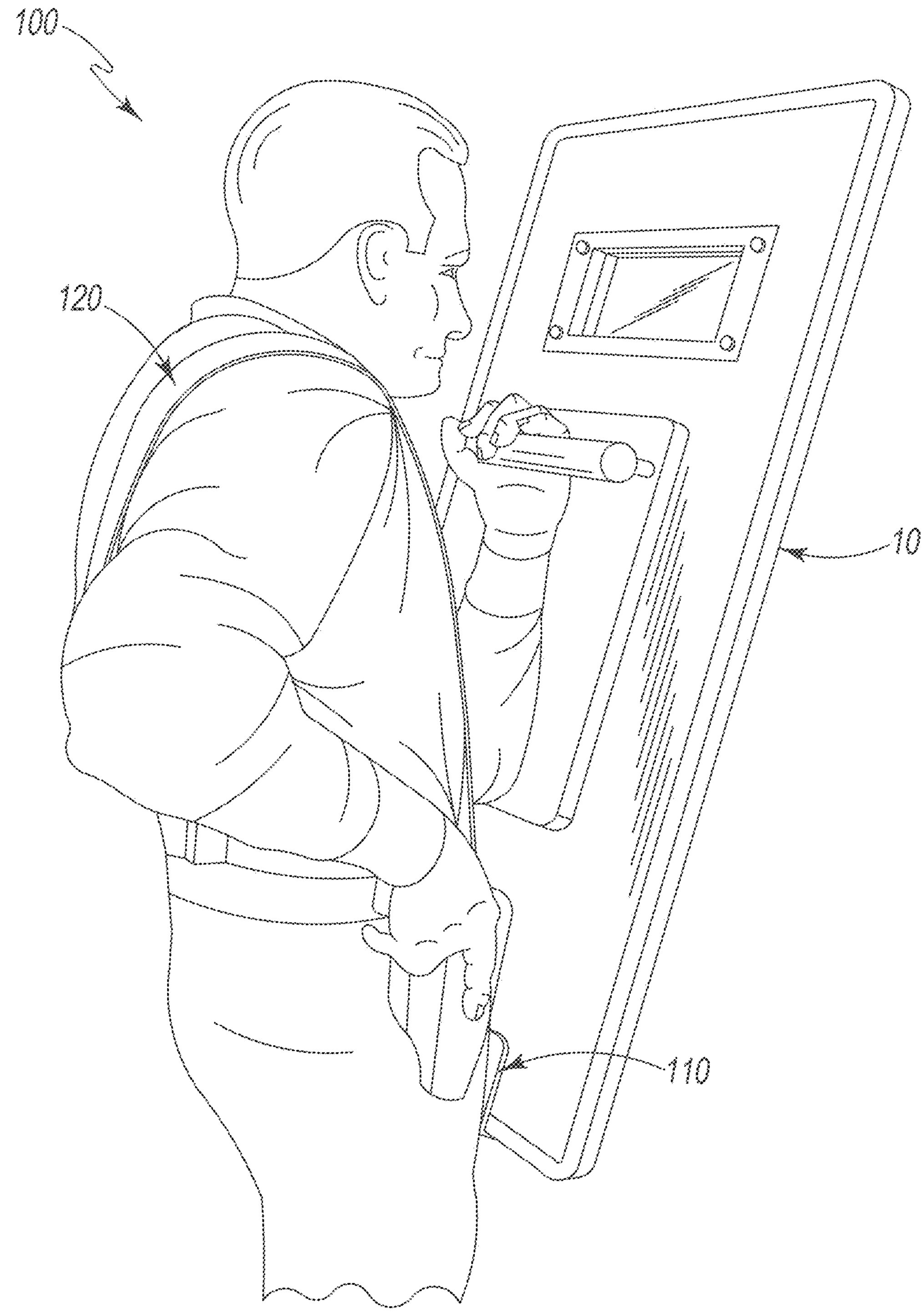


Fig. 6

BALLISTIC SHIELD HARNESS

[0001] This application claims the benefit of U.S. Provisional Patent Application, Ser. No. 63/557,004 filed Feb. 23, 2024, the disclosures of both provisional patent applications are hereby incorporated by reference.

[0002] This invention relates to a harness for supporting a ballistic shield carried by a user.

BACKGROUND AND SUMMARY OF THE INVENTION

[0003] Law enforcement or military personnel often carry ballistic shields to protect against impact weapons and incoming projectiles. These shields can weigh in excess of 50 pounds and carrying them by hand for extended periods greatly fatigues the individual personnel carrying the shield in response to threats.

[0004] The shield harness of this invention supports the weight of a ballistic shield and distributes the weight of the shield across the user's shoulders and torso. The shield harness includes a U-shaped shield saddle and a pair of adjustable shoulder straps. The shoulder straps wrap over the shoulders of the user to position and support the shield saddle and the weight of a ballistic shield in front of the user. Once donned, the user rest the bottom of the ballistic shield atop the shield saddle and holds the shield handle with one hand leaving the other hand free for other functions. The weight of the ballistic shield is supported by the shield harness, which distributes that weight across the users' shoulders rather than carried by the users' arms, thereby greatly reducing user fatigue.

[0005] The above described features and advantages, as well as others, will become more readily apparent to those of ordinary skill in the art by reference to the following detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The present invention may take form in various system and method components and arrangement of system and method components. The drawings are only for purposes of illustrating exemplary embodiments and are not to be construed as limiting the invention.

[0007] The drawings illustrate the present invention, in which:

[0008] FIG. **1** is a perspective view of an exemplary embodiment of the shield harness of this invention;

[0009] FIG. **2** is a perspective view of the shield saddle used in the shield harness of FIG. **1**;

[0010] FIG. **3** is a front perspective view of a user donning the shield harness of FIG. **1**;

[0011] FIG. **4** is another front perspective view of a user donning the shield harness of FIG. **1**;

[0012] FIG. **5** is a front side perspective view of the user carrying a ballistic shield supported by the shield harness of FIG. **1**; and

[0013] FIG. **6** is a rear side perspective view of the user carrying a ballistic shield supported by the shield harness of FIG. **1**.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0014] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific preferred embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is understood that other embodiments may be utilized and that logical, structural, mechanical, electrical, and chemical changes may be made without departing from the spirit or scope of the invention. To avoid detail not necessary to enable those skilled in the art to practice the invention, the description may omit certain information known to those skilled in the art. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims.

[0015] Referring now to the drawings, FIGS. **1-6** illustrate an exemplary embodiment of the ballistic shield harness of this invention, which is designated generally as reference numeral **100**. Shield harness **100** is designed to be worn by a user to support the weight of a conventional ballistic shield **10** carried by the user. Shield harness **100** is also designed to be worn either over or under a tactical vest. Shield harness **100** consists of U-shaped shield saddle **110** and a pair of adjustable shoulder straps **120**. Shoulder straps **120** wrap over the shoulders of the user to position and support saddle **110** and the weight of ballistic shield **10** in front of the user. In other embodiments, shield harness **100** also includes an optional waist strap that connects and supports shield saddle **110** around the user's waist. In addition, the harness of this invention can be used to aid in the carrying of anything heavy that can be fit into the saddle in other embodiments.

[0016] As best shown in FIG. **2**, shield saddle **110** takes the form of a metal plate bent into a u-shaped configuration. Saddle **110** has an integral back panel **112**, a central platform and a front panel **116**. Back panel **112** and front panel **116** are generally parallel to each other with central platform **114** extending perpendicularly there between along the length of the bottom edge of both back panel **112** and front panel **116**. Back panel **112** is taller than front panel **116** and generally configured to rest against the body of the user. The width of central platform **114** is dimensioned to receive the bottom of ballistic shield **10**. Back panel **112** has a pair of diagonal slots **113** on either side. Back panel **112** also includes a central horizontal slot **115** and a pair of vertical slots **117**. Shoulder straps **120** pass through diagonal slots **117** to secure saddle **110** to the shoulder straps. An optional waist strap **130** (FIG. **5**) can be used to securely center saddle **110** in front of the user. Waist strap passes through vertical slots **117** and wraps around the user's waist. Saddle may also have other slots and holes placed on the front and back plates to accommodate portions of ballistic shield **10** and to secure the shield therein saddle **110**, as needed.

[0017] As best shown in FIG. **1**, shoulder straps **120** take the form of two lengths of nylon webbing. The lengths of the straps can be readily adjusted and fitted to the user to position the saddle at the optimum height to support the shield over the front of the user's body. Shoulder straps **120** are constructed typically of flat woven strips or tubes of high-strength polymer materials, such as Nylon, Polyester, and Polypropylene. While polymer webbing is preferable, other traditional sling materials, such as cloth or leather straps may be used. Each shoulder strap **120** has a free end **122** and a connected end **124**. Connected ends terminate in a strap slide adjustor or tri-glide **126**. Each shoulder strap **120** also includes a D-ring **127** affixed to the strap adjacent tri-glide **126**. Tri-glide **126** and D-ring **127** are of the type available from ITW Nexus North America of Des Plaines, Illinois and allow the lengths of straps **120** to be selectively adjusted, shortening and lengthening as desired for any given user. As shown in FIG. **1**, free end **122** of each strap **120** passes through one of the diagonal slots **115** in shield saddle **110** and passes through the tri-glide **126** on the connected end **124** of the opposite shoulder strap **120**. Shoulder straps **120** cross one another and are slidably connected by a crossed connector **128**. When shoulder straps **120** are donned, cross connector **128** is positioned to overlie the user's back with shoulder straps **120** riding atop the user's shoulders and crossing across the user's back. The length of each shoulder strap **120** is adjusted by the user manually extending or retracting free ends **122** of one strap **120** through tri-glides **126** of the opposite shoulder strap.

[0018] In another embodiment, shield harness of this invention may include a separate waist strap **130** (FIG. **5**) that can be used in conjunction with shoulder straps **120**. Generally, waist strap **130** is similar in material and use as shoulder straps **120**, but is used to secure shield saddle **110** against the front of the user. Waist strap **130** passes through **11** the vertical slots **117** in shield saddle **110** and is adjustably secured by a buckle (not shown).

[0019] As shown in FIGS. **3-6**, shield harness **100** is donned by the user with shoulder straps draped over the shoulders and crossing across the user's back. Free ends **122** of each strap are manually pulled to tighten and position shield harness **110** to center in front of the user. Shield saddle **110** is positioned below the waist line to support ballistic shield **10** at the desired height relative to the user to cover the user's torso and head. Once shield harness **100** is donned and adjusted, the user lifts and rests the bottom of ballistic shield into shield saddle **110**. The user holds shield handle **12** with one hand while allowing the other hand to be free for other functions. The weight of ballistic shield **10** is supported by shield harness **100** which distributes that weight across the users' shoulders rather than carried by the user's arms, thereby greatly reducing user fatigue.

[0020] While designed and intended primarily for user to position and support an individual ballistic shield, shield harness **100** can be used by one or more users to support the weight of other items. For example, shield harness **100** can be used to support and carry litters and gurneys (not shown), in certain uses and applications. Side or end rails of a litter and gurney can be securely seated within the shield saddle **110** with shoulder straps **120** allowing the user to carry the weight of the item. Similarly, shield harness **100** can be used to support and carry large bulky containers (not shown) with exterior handles and the like. Again, the exterior handles of such containers can be seated within the shield saddle and hte shoulder straps used to carry the weight of the container. One skilled in the art, will note the advantages and versatility of the sheild harness of this invention beyond its use and application of a user to support and carry an individual ballistic sheild.

[0021] It should be apparent from the foregoing that an invention having significant advantages has been provided. While the invention is shown in only a few of its forms, it is not just limited but is susceptible to various changes and modifications without departing from the spirit thereof. The embodiment of the present invention herein described and illustrated is not intended to be exhaustive or to limit the invention to the precise form disclosed. It is presented to explain the invention so that others skilled in the art might utilize its teachings. The embodiment of the present invention may be modified within the scope of the following claims.

I claim:

1. A harness worn by a user for carrying a shield comprising:
- a shield saddle for receiving the shield; and
- a pair of shoulder straps adjustably connected to the shield saddle to be positioned over the shoulders of a user for suspending the shield saddle in front of the user;

2. The harness of claim **1** wherein the shield saddle has a U-shaped cross section and includes a back panel, front panel and a central platform connecting the back panel and front panel along the bottom edge thereof for receiving the bottom edge of the shield atop the central platform between the back panel and front panel.

3. The harness of claim **1** wherein the shield saddle has a pair of slots therein, each of the pair of slots for receiving one of the pair of shoulder straps for suspending the saddle from the pair of shoulder straps.

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